



UNIVERSITÉ DE KINSHASA
FACULTÉ DES SCIENCES ET TECHNOLOGIES



MENTION MATHÉMATIQUES, STATISTIQUE ET INFORMATIQUE

B.P. 190 KINSHASA XI

VOTRE TITRE SERA ICI

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Gradué en Sciences

Mémoire présenté et défendu en vue de l'obtention d'une maîtrise en ...

Directeur : Prof. **Prénom Nom**

Année académique 2025- 2026

VOTEX

À ADAM et ÈVE

Remerciements

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VOTRE NOM

Introduction

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Chapitre I

Généralités

Lorem ipsum, or lipsum as it is sometimes known, is dummy text used in laying out print, graphic or web designs. The passage is attributed to an unknown typesetter in the 15th century who is thought to have scrambled parts of Cicero's De Finibus [2], Bonorum et Malorum for use in a type specimen book. It usually begins with.

I.1 Rappels

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I.1.1 Définition (Problème Primal de PLSI) une généralisation de la notion de programme linéaire avec la possibilité que soit le nombre de variables est infini, soit le nombre de contraintes est infini.

Nous allons considérer le cas où le nombre des contraintes est infini. Dans ce cas, le problème primal de PLSI peut être formulé comme suit

$$(P) = \begin{cases} \min & c^T x \\ S/C & a_t^T x \geq b_t, t \in T \\ & x \geq 0 \end{cases} \quad (I.1)$$

où $c \in \mathbb{R}^n$, T est un ensemble infini arbitraire, $a_t \equiv a(t) = (a_1(t), \dots, a_n(t))$ est une application de T dans $\mathbb{R}^n$, $b_t = b(t)$ est une fonction réelle sur T .

Les notations $c^T x$ et $a_t^T x$ désignent le produit scalaire usuel de $\mathbb{R}^m$ et la notation a_t^T désigne la transposée du vecteur a_t de $\mathbb{R}^n$.

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I.2 Contexte

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I.2.1 Théorème À supposer que tout sous-système fini de A admet une solution. Alors A n'admet aucune solution si et seulement si il existe $\sum \in \mathbb{R}^n$ tel que $\sum \cdot x \geq N$ peut être déduite de A pour tout N .

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Chapitre II

Deuxième chapitre

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II.1 Rappels

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II.1.1 Définition (Problème Primal de PLSI) Un programme linéaire semi-infini est une généralisation de la notion de programme linéaire avec la possibilité que ou bien le nombre de variables est infini, ou bien le nombre de contraintes est infini.

Nous allons considérer le cas où le nombre des contraintes est infini. Dans ce cas, le problème primal de PLSI peut être formulé comme suit

$$(P) = \begin{cases} \min & c^T x \\ S/C & a_t^T x \geq b_t, t \in T \\ & x \geq 0 \end{cases} \quad (II.1)$$

où $c \in \mathbb{R}^n$, T est un ensemble infini arbitraire, $a_t \equiv a(t) = (a_1(t), \dots, a_n(t))$ est une application de T dans $\mathbb{R}^n$, $b_t = b(t)$ est une fonction réelle sur T .

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Conclusion

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